# Assignment 3: Report

## Deep Generative Models: VAEs for Conditional Face Expression Generation

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## Solution Description

We implement a conditional variational autoencoder (CVAE) using TensorFlow/Keras to generate face expression images conditioned on one of four classes: Happy, Sad, Surprised, and Mad.

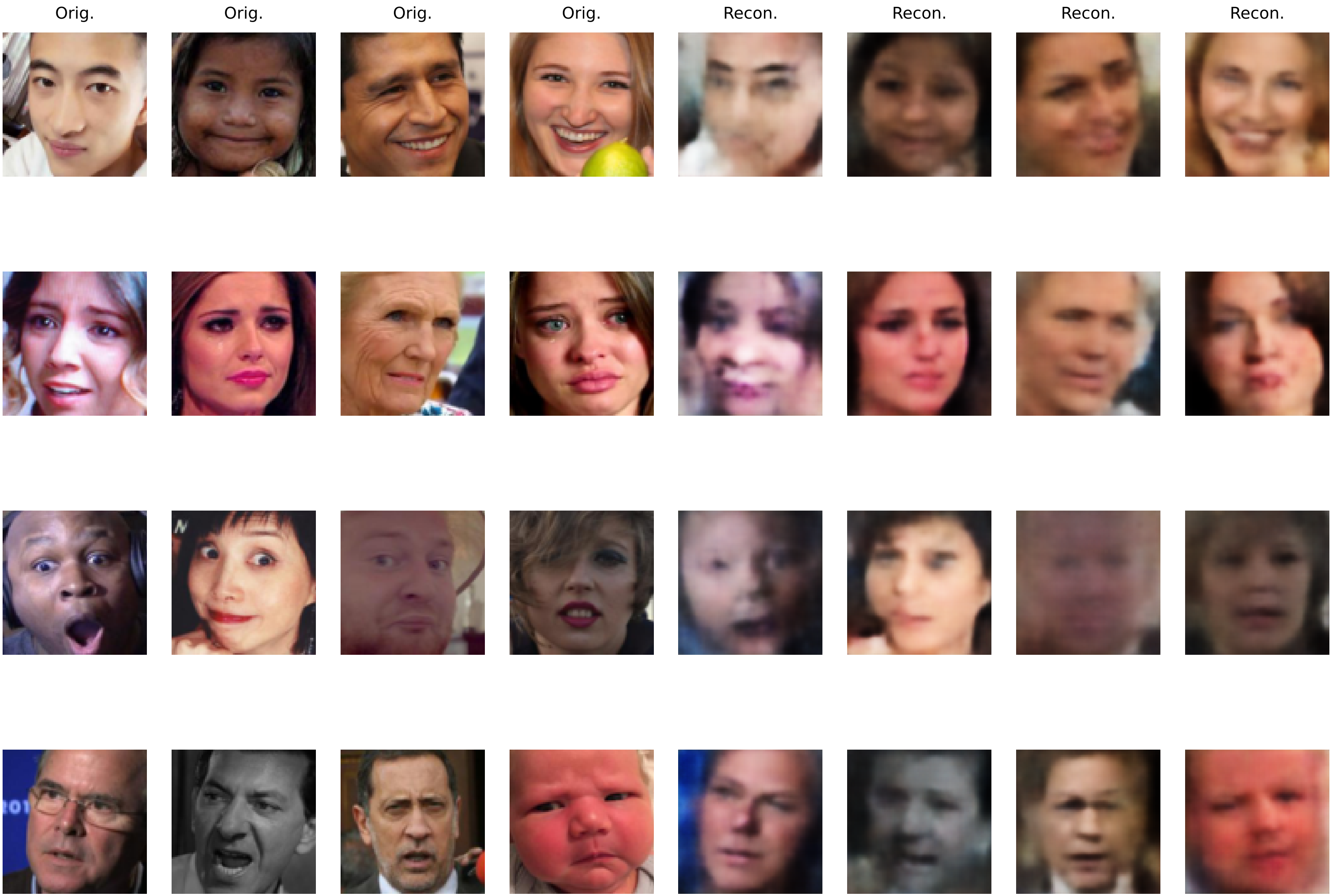
Architecture. The encoder receives a 112x112x3 image concatenated with a spatially broadcast learned label map (Dense projection of the one-hot label to 112x112x1), then passes through four convolutional layers (32, 64, 128, 256 filters, stride 2, 4x4 kernels) with batch normalisation and ReLU, reducing to 7x7. The flattened features are concatenated with the one-hot label again and mapped via a dense layer (512 units) to mean and log-variance vectors in a 128-dimensional latent space. Sampling uses the reparameterisation trick: z = mu + exp(0.5 * log_var) * epsilon, epsilon ~ N(0, I). The decoder concatenates z with the one-hot label, maps through dense layers to a 7x7x256 feature map, then applies three transposed-convolution layers (128, 64, 32 filters) with a label injection at every spatial level: a Dense projection of the label to a sz x sz x 1 map is concatenated with the feature maps after each upsampling. A final transposed-convolution layer outputs the 112x112x3 image with sigmoid activation.

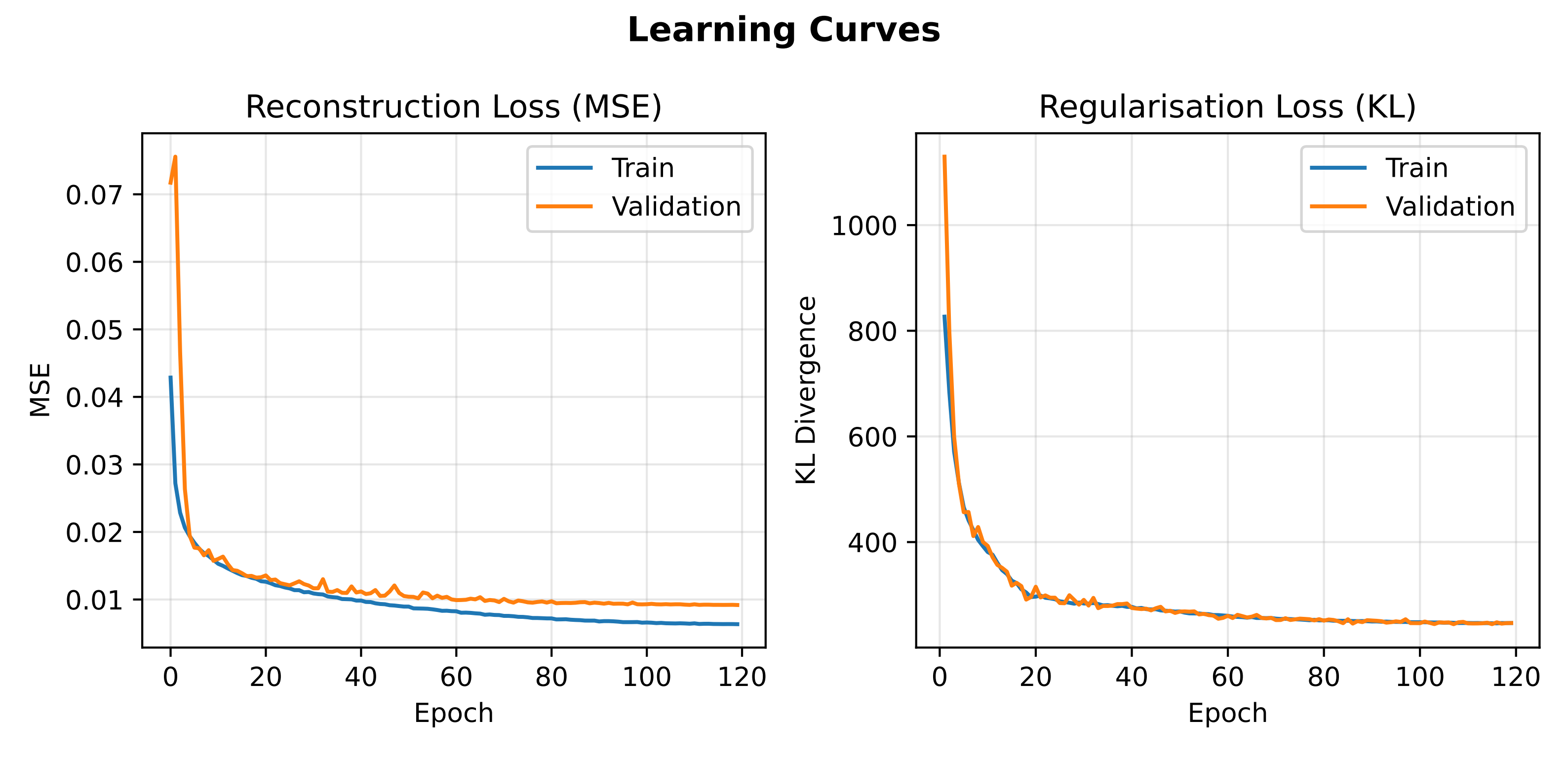
Loss function. The total loss combines MSE reconstruction (summed over pixels per sample) with an analytical KL divergence: KL = -0.5 * sum(1 + log_var - mu^2 - exp(log_var)). We use beta = 0.15 with linear KL annealing over 20 epochs to prevent posterior collapse.

Training. Adam with cosine learning rate decay (1e-3 to 1e-5) over 120 epochs, batch size 64 on GPU, with random horizontal flip augmentation. The best model (by validation MSE) is checkpointed. The reduced beta allows the latent space to encode richer class-specific information, improving interpolation quality.

Challenges. Balancing reconstruction vs. latent regularity was key: beta=1.0 caused posterior collapse and washed-out interpolations. Multi-level label injection in the decoder significantly improved class-conditional generation. KL annealing with reduced beta resolved these issues.

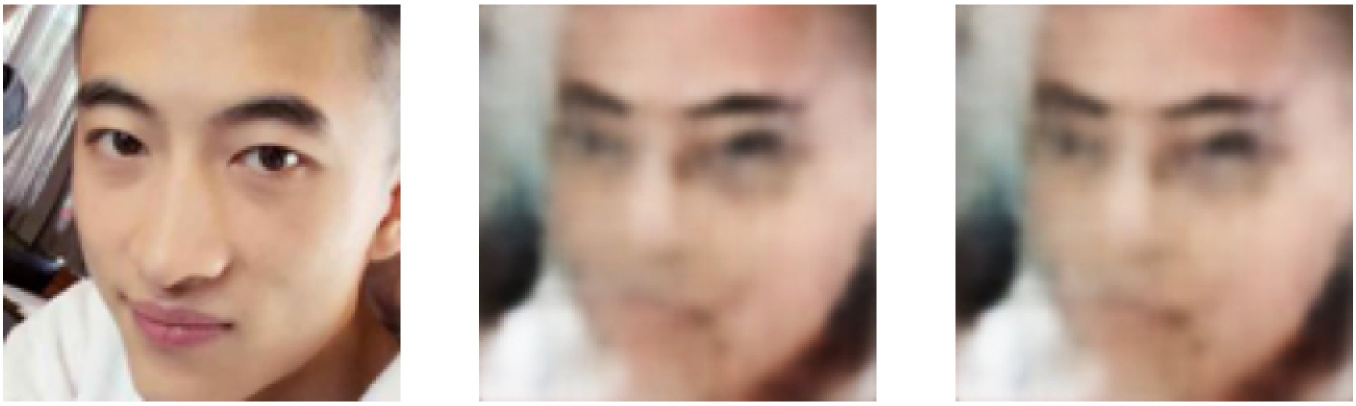
**Sample Images: Original (left 4) vs Reconstructed (right 4)**




Learning Curves
Reconstruction Loss (MSE)
MSE
Epoch
Train
Validation
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
20
40
60
80
100
120
Regularisation Loss (KL)
KL Divergence
Epoch
Train
Validation
1000
800
600
400
0
20
40
60
80
100
120

# Task 2: Latent Space Interpolation

Original (Happy) | Recon (Happy cond.) | Same latent, Sad cond.



Task 2.1: A Happy image is encoded and its latent vector is decoded with the Sad class-conditioning signal. The generated image does not look like a sad version of the original person. The latent vector was encoded under Happy conditioning, so it carries features that belong to the Happy class. Decoding it with Sad conditioning makes the decoder treat that latent as if it came from Sad data, so the output looks like a Sad face but loses the original identity. Both the conditioning signal and the latent code affect the result, so swapping only the condition is not enough to keep the identity across classes.

## Latent Interpolation: Happy mean → Sad mean (Sad conditioning)

α=0.0 α=0.1 α=0.2 α=0.3 α=0.4 α=0.6 α=0.7 α=0.8 α=0.9 α=1.0

Task 2.2: We compute the mean latent vector (mu) for all Happy training images and all Sad training images. We then linearly interpolate between these two mean vectors in 10 equally spaced steps (alpha = 0.0 to 1.0), decoding each interpolated vector with the Sad conditioning signal.

The interpolations give plausible transitions. At alpha=0 (Happy mean) the generated face shows a slight smile, and at alpha=1 (Sad mean) the expression becomes more neutral or downturned. The change is gradual and there are no sudden jumps between steps. The faces stay blurry throughout, which is typical for a VAE trained with MSE loss since it tends to average over possible outputs instead of producing sharp images. The faces shift expression gradually rather than jumping between distinct classes, which suggests the CVAE has learned a fairly continuous latent space where the expression can be moved along a direction.